

The Lab-Book concept for the CREATE Modules

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- The CREATE modules are designed in the context of existing graduate programs.
- They may thus be used to supplement a graduate program or as part of courses for credit (typically reading type courses).
- Different modules have a different balance between the types of material students are asked to digest. Some may include a substantial code component.
- Students also bring different motivations and goals to the modules.
- Based on my experience teaching during Covid students appreciate having some choice in their evaluations, and the agency gained often improves the quality of work.
- The standard “work time” for a course should be 8 hours in a week, including lectures and rereading slides, looking for references and so on.
- Remember that quantitative methods need work by hand, but it cannot be all about lengthy algebra. An essential portion of the CREATE is communication.

Lab book structure

60% exercises and simple tasks

30% mid-sized problems requiring some set up and description

10% one larger task requiring reading something not explicitly assigned, or writing new code

Small tasks

- Can include a time tracker of readings
- Can include a diary of code runs and small changes (parameter value changes, new plots)
- For theory textbook problems may be found and included. I have typically asked students to list problems done and type up a subset.

Medium tasks

- Should start with a clear articulation of the purpose of the task.
- For code changes include relevant changes.
- For theory, can use textbook problems that build.
- The task should be typeset and proofread.

Larger Task

- Communicate with the module instructor and record the discussion time and the main gist of what was agreed to.
- Include the reference for the novel task. Note this could involve traditional sources (an article) or non-traditional sources (implementing software a more senior student uses).
- Agree on a deliverable (typically 2/3 of write-up, code, talk).
- Talks should be conference length (12 minutes) to reduce bloat

The larger task will take more effort than the other portions!

- The Perturbation Theory portion of this module has lots of possible sources.
- I have included sample quizzes and a take-home from the “full” version of this material as part of the AMATH 732 course at UW.
- I have also included some sample quizzes from the AMATH 463/663 fluid mechanics course at UW.
- These are good for the small and medium parts.
- The fluid mechanics portions are a bit trickier and you could consider weighing the exercise more to perturbation theory and perhaps choosing a fluid mechanics related long exercise.